

Ramkumar Natarajan

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EDUCATION

Carnegie Mellon University

PhD in Robotics

Advisors: Maxim Likhachev and Howie Choset

Pittsburgh, PA

Aug 2018 – Ongoing

Worcester Polytechnic Institute

MS in Robotics (Ranked 1st in class)

Worcester, MA

Aug 2015 – June 2017

SASTRA Unviersity

BS in Electrical Engineering

Thanjavur, India

Aug 2011 – May 2015

WORK EXPERIENCE

Graduate Research Assistant

Search Based Planning Lab and Biorobotics Lab at The Robotics Institute, CMU

Aug 2018 – Present

Pittsburgh, PA

- PhD candidate developing motion planning algorithms for highly dynamic tasks

Senior Robotics Researcher

RobotWits LLC (Acquired by Waymo)

Jun 2017 – Jul 2019

Pittsburgh, PA

- As a part of an early startup, I was solely responsible for the company's entire technical development for months together. I established the operational procedures, development processes, and the hiring pipeline. Subsequently, I facilitated the hiring of four new engineers and mentored them on system design and development.
- Led the design and development of the multi-tier motion planning stack containing low-level high-speed planning, high-level route planning, and deliberative planning under uncertainty for seamless integration with perception, mapping, and control subsystem as well as the simulator through a middleware.
- Implemented the core algorithmic components of motion planning stack while maintaining soft real-time constraints on the application layer.
- Led the integration of the motion planning stack with other autonomy subsystems on physical vehicles for various clients and delivered multiple projects from concept to completion.
- Designed the fail-operational architecture for redundancy in the planning subsystem of self-driving vehicles.

Robotics Engineer

Bossanova Robotics

May 2016 – May 2017

Pittsburgh, PA

- Extended Viola Jones rapid object detection algorithm to multi-channel images (color, depth, and intensity).
- Incorporated graph reductionist and tree decomposition methods to parallelize filtering and smoothing for SLAM.

INTERNSHIPS

Research Intern

Near Earth Autonomy

May 2021 – Aug 2021

Pittsburgh, PA

- Planning under uncertainty for autonomous landing of an unmanned aerial vehicle on a naval destroyer

Visiting Researcher

Indian Institute of Technology(IIT) Madras

Aug. 2014 – May 2015

Chennai, India

- Real-time navigation for Pioneer P3-DX mobile robot using lifelong SLAM

PUBLICATIONS

- [1] Ramkumar Natarajan, Chaoqi Liu, Howie Choset, and Maxim Likhachev. Implicit graph search for planning on graphs of convex sets. In *Robotics: Science and Systems (RSS)*, 2024.
- [2] Ramkumar Natarajan, Shohin Mukherjee, Howie Choset, and Maxim Likhachev. **PINSAT: Parallelized Interleaving of Graph Search and Trajectory Optimization for Kinodynamic Motion Planning.** Under Review at *IROS 2024* - *arXiv preprint arXiv:2401.08948*, 2024.

- [3] Ramkumar Natarajan, Garrison L. H. Johnston, Nabil Simaan, Maxim Likhachev, and Howie Choset. **Long Horizon Planning through Contact using Discrete Search and Continuous Optimization**. *Under Review at IEEE Transactions on Robotics*.
- [4] Ramkumar Natarajan*, Hanlan Yang*, Qintong Xie, Yash Oza, Manash Pratim Das, Fahad Islam, Muhammad Suhail Saleem, Howie Choset, and Maxim Likhachev. **Preprocessing-based Kinodynamic Motion Planning Framework for Intercepting Projectiles using a Robot Manipulator**. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024.
- [5] Ramkumar Natarajan, Garrison LH Johnston, Nabil Simaan, Maxim Likhachev, and Howie Choset. **Torque-limited Manipulation Planning through Contact by Interleaving Graph Search and Trajectory Optimization**. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 8148–8154. IEEE, 2023.
- [6] Ramkumar Natarajan, Howie Choset, and Maxim Likhachev. **Interleaving Graph Search and Trajectory Optimization for Aggressive Quadrotor Flight**. *IEEE Robotics and Automation Letters*, 6(3):5357–5364, 2021.
- [7] Ramkumar Natarajan, Muhammad Saleem, Sandip Aine, Maxim Likhachev, and Howie Choset. **A-MHA*: Anytime Multi-Heuristic A***. In *Proceedings of the International Symposium on Combinatorial Search*, volume 10, pages 192–193, 2019.
- [8] Ramkumar Natarajan and Michael A Gennert. **Efficient Factor Graph Fusion for Multi-robot Mapping and Beyond**. In *2018 21st International Conference on Information Fusion (FUSION)*, pages 1137–1145. IEEE, 2018.
- [9] Siddharthan Rajasekaran*, Ramkumar Natarajan*, and Jonathan D Taylor. **Towards Planning and Control of Hybrid Systems with Limit Cycle using LQR Trees**. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5196–5203. IEEE, 2017.

PATENTS

- [1] Jonathan Davis Taylor and Ramkumar Natarajan. **Color Haar Classifier for Retail Shelf Label Detection**. Google Patents, June 13 2019. US Patent App. 16/219,238.

TECHNICAL REPORTS

- [1] Sri Ramana Sekharan, Ramkumar Natarajan, and Siddharthan Rajasekaran. **Transfer from multiple linear predictive state representations (PSR)**. *arXiv preprint arXiv:1702.02184*, 2017.

THESIS

- [1] Ramkumar Natarajan. **Efficient Factor Graph Fusion for Multi-robot Mapping**, 2017.

TECHNICAL SKILLS

Languages: C/C++, Python, Matlab, Julia, Verilog

Developer Tools: ROS, Eigen, Boost, PyTorch, OpenCV, Git, Docker, SolidWorks, Arduino

INVITED TALKS

Implicit Graph Search for Planning on Graphs of Convex Sets

Robot Locomotion Group, MIT

Nov 2023

Cambridge, MA

Use of Topology in Optimal Motion Planning

Workshop on Topological Methods in Motion Planning

ICRA 2019

Montreal, Canada

TEACHING

16-745 Optimal Control and Reinforcement Learning

Spring 2021

Teaching Assistant for course taught by Chris Atkeson

The Robotics Institute, CMU

16-782 Planning and Decision-making in Robotics

Fall 2020

Teaching Assistant for course taught by Maxim Likhachev

The Robotics Institute, CMU

ECE 3829: Advanced Digital System Design using FPGAs

Spring 2016

Teaching Assistant

WPI

SCHOLASTIC ACHIEVEMENTS

Ranked 1st in the MS in Robotics Engineering in a class of 48 students.

2017

Worcester Polytechnic Institute (WPI)

Worcester, MA

Junior Research Fellowship

2013

Center for Artificial Intelligence & Robotics - Defense Research and Development Organization (CAIR - DRDO) India

REFERENCES

Maxim Likhachev

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Professor, The Robotics Institute, CMU

Howie Choset

choset@cs.cmu.edu

Professor, The Robotics Institute, CMU